

Asteria Boutique Hotel - Aegina, Greece

This document outlines the data analysis project for Asteria Boutique Hotel on the island of Aegina, Greece. The analysis covers the period from 1 January 2024 to 31 December 2025. Two CSV files were provided: *hotel_bookings_raw.csv*, containing 3,015 rows, and *daily_room_inventory.csv*, containing 2,193 rows.

The project is structured into three stages:

1. **Data quality check and cleaning:** identifying data quality issues and preparing the datasets for analysis.
2. **Data analysis:** exploring the cleaned data to evaluate hotel performance and identify patterns and trends.
3. Findings and recommendations: addressing stakeholders' questions, explaining the key insights, and presenting recommendations supported by the analysis.

Data Quality

(*hotel_bookings_raw.csv*)

DQ_001

3,000 unique booking codes were found across a total of 3,015 rows. Therefore, there are 15 additional occurrences of booking codes. The check does not prove that the corresponding rows are identical across all columns.

Query:

```
SELECT
  DISTINCT booking_id
FROM
  `Asteria-Boutique-Hotel.hotel_bookings_raw`
```

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	booking_id					
1	BK001115					
2	BK001528					
3	BK000016					
4	BK002706					
5	BK001858					
6	BK001527					
7	BK001077					
8	BK000344					
9	BK000598					
10	BK001009					
11	BK000291					
12	BK000694					
13	BK002290					
14	BK002053					
15	BK002825					
16	BK001089					
17	BK000997					
18	BK000920					
19	BK001449					
20	BK000752					
21	BK001786					
22	BK000687					
23	BK002929					

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DQ_002

15 duplicate records were found. 15 codes appear twice — 30 lines in total, 15 additional occurrences.

Query:

```
SELECT
  booking_id, COUNT(booking_id) AS booking_id_duplicates
FROM
  `Asteria-Boutique-Hotel.hotel_bookings_raw`
GROUP BY
  booking_id
HAVING
  booking_id_duplicates > 1
```

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	booking_id	booking_id_dupli...				
1	BK001469	2				
2	BK000132	2				
3	BK001364	2				
4	BK002058	2				
5	BK000938	2				
6	BK000818	2				
7	BK000178	2				
8	BK000537	2				
9	BK002619	2				
10	BK001140	2				
11	BK001003	2				
12	BK000923	2				
13	BK000881	2				
14	BK002842	2				
15	BK000548	2				

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DQ_003

Check: check-in after check-out — 0 records — no problem detected.

Query:

```
SELECT
    booking_id, booking_date, check_in, check_out
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    check_in > check_out
```

DQ_004

Check: **booking_date** after **check_out** — 0 entries — no problem found.

Query:

```
SELECT
    booking_id, booking_date, check_in, check_out
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    booking_date > check_out
```

DQ_005

No values outside the 0–100 range were found in the **discount_pct** column (i.e., values less than 0 or greater than 100).

Query:

```
SELECT
    discount_pct
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    discount_pct < 0 OR discount_pct > 100
```

DQ_006

No NULLs found in **booking_id**.

Query:

```
SELECT
    booking_id
FROM
    `Asteria-Boutique_Hotel.hotel_bookings_raw`
WHERE
    booking_id IS NULL
```

DQ_007

Category inconsistency found in the following columns:

booking_channel
trip_purpose
guest_country

Query 1:

```
SELECT
    DISTINCT booking_channel
FROM
    `Asteria-Boutique_Hotel.hotel_bookings_raw`
```

Job information		Results	Visualization	JSON	Execution dt
Row	booking_channel ▼				
1	booking.com				
2	direct				
3	travel agency				
4	Booking.com				
5	Direct				
6	Expedia				
7	Travel Agency				
8	expedia				

Query 2:

```
SELECT
    DISTINCT
        trip_purpose
FROM
    `Asteria-Boutique_Hotel.hotel_bookings_raw`
```

Job information		Results	Visualization	JSON	Execu
Row	trip_purpose ▼				
1	Leisure				
2	Business				
3	leisure				
4	business				

Query 3:

```
SELECT
    DISTINCT guest_country
FROM
    `Asteria-Boutique_Hotel.hotel_bookings_raw`
```

Job information		Results	Visualization	JSON	Exec
Row	guest_country				
1	GR				
2	ES				
3	IT				
4	GB				
5	NL				
6	FR				
7	DE				
8	US				
9	nl				
10	null				
11	gr				
12	gb				

*No inconsistencies were detected in the **booking_status**, **room_type**, and **payment_method** columns.

DQ_008

Maximum observed number of adults/children (2) in the **adults** column.

Query:

```
SELECT
    booking_id, MAX(adults) as max_adults
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
GROUP BY
    booking_id
ORDER BY
    max_adults DESC
LIMIT
    5
```

Job information		Results	Visualization	JSON
Row	booking_id	max_adults		
1	BK001527	2		
2	BK001528	2		
3	BK000016	2		
4	BK000598	2		
5	BK001858	2		

DQ_009

Maximum observed number of children: from the **children** column.

Query:

```
SELECT
    booking_id, MAX(children) as max_children
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
GROUP BY
    booking_id
ORDER BY
    max_children DESC
LIMIT
    5
```

Job information		Results	Visualization	JSON	Execution
Row	booking_id	max_children ▼			
1	BK002282	2			
2	BK001786	2			
3	BK002890	2			
4	BK001526	2			
5	BK001180	2			

DQ_010

No negative values were found in the **adults** and **children** columns.

Query:

```
SELECT
    booking_id, adults, children
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    adults < 0 OR children < 0
```

DQ_011

No zero values were found in the **adults** column.

Query:

```
SELECT
    booking_id, adults
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    adults = 0
```

DQ_012

Values containing two names, as well as NULL values, were found in the country-of-origin column (**guest_country**).

Query:

```
SELECT
    DISTINCT guest_country
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
```

Job information		Results	Visualization	JSON	Execution det
Row	guest_country				
1	GR				
2	ES				
3	IT				
4	GB				
5	NL				
6	FR				
7	DE				
8	US				
9	nl				
10	null				
11	gr				
12	gb				

DQ 013

No negative values were found in the columns:

nightly_rate_eur,
room_total_eur,
extras_total_eur,
commission_eur,
booking_value_eur

Query:

```
SELECT
    nightly_rate_eur,
    room_total_eur,
    extras_total_eur,
    commission_eur, booking_value_eur
FROM
    `AsteriaBoutiqueHotel.hotel_bookings_raw`
WHERE
    nightly_rate_eur < 0 OR
    room_total_eur < 0 OR
    extras_total_eur < 0 OR
    commission_eur < 0 OR
    booking_value_eur < 0
```

DQ 014

In the **review_score** column, 15 values were found that fell outside the rating scale.

Query:

```
SELECT
    booking_id, review_score
FROM
    `AsteriaBoutiqueHotel.hotel_bookings_raw`
WHERE review_score < 1 OR review_score > 10
```

Job information		Results	Visualization	JSON	Exec
Row	booking_id	review_score			
1	BK002044	-1			
2	BK000102	-1			
3	BK000175	11			
4	BK001387	-1			
5	BK000225	-1			
6	BK000064	-1			
7	BK000608	-1			
8	BK000365	11			
9	BK001372	11			
10	BK001610	11			
11	BK002635	11			
12	BK001408	11			
13	BK000300	-1			
14	BK001538	11			
15	BK000305	11			

DQ 015

No NULL records were found in the columns:

booking_date, check_in, check_out, adults, children, nightly_rate_eur, room_total_eur, extras_total_eur,
commission_eur, booking_value_eur

Query:

```
SELECT
    booking_id,
    booking_date,
    check_in,
    check_out,
    adults,
    children,
    nightly_rate_eur,
    room_total_eur,
    extras_total_eur,
    commission_eur,
    booking_value_eur
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    booking_date IS NULL OR
    check_in IS NULL OR
    check_out IS NULL OR
    adults IS NULL OR
    children IS NULL OR
    nightly_rate_eur IS NULL OR
    room_total_eur IS NULL OR
    extras_total_eur IS NULL OR
    commission_eur IS NULL OR
    booking_value_eur IS NULL
```

DQ 016

15 records with a difference of +€25 between the sum of room extras and the total value. The initial check returned 73 lines, but 58 were excluded after rounding to two decimal places because the differences were negligible.

Query:

```
SELECT
    booking_id,
    room_total_eur,
    extras_total_eur,
    booking_value_eur,
    room_total_eur + extras_total_eur - booking_value_eur as difference_eur
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    ROUND (room_total_eur + extras_total_eur - booking_value_eur, 2) <> 0
ORDER BY
    difference_eur DESC
```

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	booking_id	room_total_eur	extras_total_eur	booking_value_eur	difference_eur	
1	BK000920	332.48	50	357.48	25.0	
2	BK001921	343.68	35	353.68	25.0	
3	BK000084	229.95	50	254.95	25.0	
4	BK002603	876.52	35	886.52	25.0	
5	BK002024	707.64	0	682.64	25.0	
6	BK000706	524.15	35	534.15	25.0	
7	BK002719	648.96	20	643.96	25.0	
8	BK000846	178.9	10	163.9	25.0	
9	BK000495	646.24	35	656.24	25.0	
10	BK001735	1096.08	35	1106.08	25.0	
11	BK000140	297.19	50	322.19	25.0	
12	BK001788	148.52	20	143.52	25.0	
13	BK000968	920.23	0	895.23	25.0	
14	BK001454	606.2	0	581.2	25.0	
15	BK000115	513.65	0	488.65	25.0	

DQ_017

Compared recorded room totals with expected amounts calculated from stay duration, nightly rate, and discount percentage. Identified 15 records with a €25 discrepancy and 49 records with a €0.01 discrepancy. The €25 discrepancies were corrected during data cleaning. The €0.01 differences were accepted as negligible rounding discrepancies and left unchanged.

Query:

```

SELECT
    booking_id,
    room_total_eur,
    ROUND(DATE_DIFF(check_out, check_in, DAY) * nightly_rate_eur * (1 - discount_pct/100),
2) as expected_room_total_eur,
    ROUND(DATE_DIFF(check_out, check_in, DAY) * nightly_rate_eur * (1 - discount_pct/100),
2) - room_total_eur AS difference_total_eur
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    ROUND(DATE_DIFF(check_out, check_in, DAY) * nightly_rate_eur * (1 - discount_pct/100),
2) <> ROUND(room_total_eur, 2)
ORDER BY
    difference_total_eur ASC

```

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	booking_id	room_total_eur	expected_room_t...	difference_total...		
1	BK001788	148.52	123.52	-25.0000000000...		
2	BK002024	707.64	682.64	-25.0		
3	BK000115	513.65	488.65	-25.0		
4	BK001921	343.68	318.68	-25.0		
5	BK001454	606.2	581.2	-25.0		
6	BK000084	229.95	204.95	-25.0		
7	BK000968	920.23	895.23	-25.0		
8	BK000140	297.19	272.19	-25.0		
9	BK001735	1096.08	1071.08	-25.0		
10	BK002603	876.52	851.52	-25.0		
11	BK000495	646.24	621.24	-25.0		
12	BK000846	178.9	153.9	-25.0		
13	BK000920	332.48	307.48	-25.0		
14	BK002719	648.96	623.96	-25.0		
15	BK000706	524.15	499.15	-25.0		
16	BK000707	2089.15	2089.14	-0.010000000000...		
17	BK000440	460.85	460.84	-0.010000000000...		
18	BK001930	352.22	352.21	-0.010000000000...		
19	BK001549	293.72	293.71	-0.010000000000...		
20	BK002026	381.05	381.04	-0.009999999999...		
21	BK000641	259.64	259.63	-0.009999999999...		
22	BK000564	869.54	869.53	-0.009999999999...		
23	BK000794	301.82	301.81	-0.009999999999...		

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DQ_018

A corresponding check on the accuracy of commissions and final amounts identified 75 lines with 73 unique booking IDs: 11 lines with differences of €2.50–€4.50—linked to the €25 discrepancies—and 64 lines with a one-cent variance.

Query:

```
SELECT
    booking_id,
    commission_eur,
    ROUND(room_total_eur * commission_pct / 100, 2) AS expected_commission,
    (ROUND(room_total_eur * commission_pct / 100, 2) - commission_eur) as
difference_commission_eur
FROM
    `Astertia-Boutique-Hotel.hotel_bookings_raw`
WHERE
    commission_eur <> ROUND(room_total_eur * commission_pct / 100, 2)
ORDER BY
    difference_commission_eur DESC
```

Job information					
Results					
Visualization					
JSON					
Execution details					
Execution graph					
Row	booking_id	commission_eur	expected_commi...	difference_comm...	
1	BK000706	89.85	94.35	4.5	
2	BK000115	87.96	92.46	4.5	
3	BK002719	112.31	116.81	4.5	
4	BK000920	46.12	49.87	3.75	
5	BK001921	47.8	51.55	3.75	
6	BK000846	23.09	26.84	3.75	
7	BK002024	102.4	106.15	3.75	
8	BK000140	27.22	29.72	2.5	
9	BK002603	85.15	87.65	2.5	
10	BK001735	107.11	109.61	2.5	
11	BK001788	12.35	14.85	2.5	
12	BK002056	97.1	97.09	-0.009999999999...	
13	BK000485	103.24	103.23	-0.009999999999...	
14	BK002247	74.21	74.2	-0.009999999999...	
15	BK001938	230.6	230.59	-0.009999999999...	
16	BK002310	119.99	119.98	-0.009999999999...	
17	BK000825	148.07	148.06	-0.009999999999...	
18	BK000403	236.54	236.53	-0.009999999999...	
19	BK000679	150.89	150.88	-0.009999999999...	
20	BK002619	82.82	82.81	-0.009999999999...	
21	BK000196	125.24	125.23	-0.009999999999...	
22	BK001400	131.57	131.56	-0.009999999999...	
23	BK002619	82.82	82.81	-0.009999999999...	

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DQ_019

Three records were found with cancelled bookings (**booking_status**) but with a review score (**review_score**). Linked to the erroneous findings of **DQ_014**.

Query:

```
SELECT
    booking_id, review_score, booking_status
FROM
    `Astertia-Boutique-Hotel.hotel_bookings_raw`
WHERE
    review_score IS NOT NULL AND booking_status <> "Completed"
```

Job information		Results	Visualization	JSON	Execution details
Row	booking_id	review_score	booking_status		
1	BK000300	-1	Cancelled		
2	BK001538	11	Cancelled		
3	BK000305	11	Cancelled		

DQ_020

No Single bookings were found with a number of **adults** other than 1 or a number of **children** other than 0.

Query:

```
SELECT
    booking_id, room_type, adults, children
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    room_type = "Single" AND (adults <> 1 OR children <> 0)
```

DQ_021

No "Cancelled" or "No-show" bookings were found with an **extras_total_eur** value other than 0.

Query:

```
SELECT
    booking_id, booking_status, extras_total_eur
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    extras_total_eur <> 0 AND (booking_status = "Cancelled" OR booking_status = "No-show")
```

DQ_022

No direct bookings with a commission rate other than 0% were detected. The check accounted for case variations and leading/trailing spaces.

Query:

```
SELECT
    booking_id, LOWER(TRIM(booking_channel)), commission_pct
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    commission_pct <> 0 AND LOWER(TRIM(booking_channel)) = "direct"
```

DQ_023

No booking dates (**booking_date**) were found that were on or after the check-in date (**check_in**).

Query:

```
SELECT
    booking_id, booking_date, check_in
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    booking_date >= check_in
```

DQ_024

No lines were found with commission rates (**commission_pct**) other than 15%, 18%, and 10% for the Booking.com, Expedia, and Travel Agency platforms (**booking_channel**), respectively.

Query1:

```
SELECT
    booking_id, commission_eur, commission_pct, LOWER(TRIM(booking_channel))
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    commission_pct <> 15 AND LOWER(TRIM(booking_channel)) = "booking.com"
```

Query 2:

```
SELECT
    booking_id, commission_eur, commission_pct, LOWER(TRIM(booking_channel))
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    commission_pct <> 18 AND LOWER(TRIM(booking_channel)) = "expedia"
```

Query 3:

```
SELECT
    booking_id, commission_eur, commission_pct, LOWER(TRIM(booking_channel))
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    commission_pct <> 10 AND LOWER(TRIM(booking_channel)) = "travel agency"
```

DQ_025

No NULLs were found in the **discount_pct** and **commission_pct** columns.

Query:

```
SELECT
    booking_id, discount_pct, commission_pct
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_raw`
WHERE
    discount_pct IS NULL OR commission_pct IS NULL
```

DQ_026

Consistency check between **booking_status** and **payment_method**: no "Cancelled" or "No-show" bookings with "Bank Transfer," "Card," or "Cash" payment methods were found. The check returned 0 records.

Query:

```
SELECT
    booking_id, LOWER(TRIM(booking_status)), payment_method
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    (
        payment_method = "Bank Transfer"
        OR payment_method = "Card"
        OR payment_method = "Cash"
    )
    AND
    (
        LOWER(TRIM(booking_status)) = "cancelled"
        OR LOWER(TRIM(booking_status)) = "no-show"
    )
```

DQ_027

Booking status and payment consistency check: no completed bookings (**booking_status** = Completed) with a "not charged" indicator (**payment_method** = Not charged) were found. The check returned 0 records.

Query:

```
SELECT
    booking_id, LOWER(TRIM(booking_status)), payment_method
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    LOWER(TRIM(booking_status)) = "completed"
    AND
    payment_method = "Not charged"
```

DQ_028

No records were found where the **check_in** and **check_out** columns have the same date.

Query:

```
SELECT
    booking_id, check_in, check_out
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    check_in = check_out
```

DQ_029

8 rows with NULL were found in the **guest_country** column.

Query:

```
SELECT
    COUNT(*) AS missing_countries_rows
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    guest_country IS NULL
```

Job information			Results	Visualizat
Row	missing_countrie...			
1	8			

DQ_030

The **booking_ID** for the 8 lines shown in **DQ-029** are as follows:

Query:

```
SELECT
    booking_id, guest_country
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    guest_country IS NULL
```

Job information			Results	Visualization	JSON	Execut
Row	booking_id	guest_country				
1	BK001426	null				
2	BK000945	null				
3	BK000670	null				
4	BK000404	null				
5	BK000824	null				
6	BK002689	null				
7	BK000020	null				
8	BK002760	null				

DQ_031

No NULLs were found in the following columns:

booking_id, guest_id, booking_status, booking_channel, room_type, trip_purpose, payment_method.

Query:

```
SELECT
    booking_id, guest_id, booking_status, booking_channel, room_type,
    trip_purpose, payment_method
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    booking_id IS NULL OR
    guest_id IS NULL OR
    booking_status IS NULL OR
    booking_channel IS NULL OR
    room_type IS NULL OR
    trip_purpose IS NULL OR
    payment_method IS NULL
```

DQ_032

No empty cells were found in the following columns:

***booking_id,
guest_id,
booking_status,
booking_channel,
room_type,
trip_purpose,
payment_method,
guest_country***

Query:

```
SELECT
    booking_id,
    guest_id,
    booking_status,
    booking_channel,
    room_type,
    trip_purpose,
    payment_method,
    guest_country
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
WHERE
    TRIM(booking_id) = "" OR
    TRIM(guest_id) = "" OR
    TRIM(booking_status) = "" OR
    TRIM(booking_channel) = "" OR
    TRIM(room_type) = "" OR
    TRIM(trip_purpose) = "" OR
    TRIM(payment_method) = "" OR
    TRIM(guest_country) = ""
```

Data Quality

(daily_room_inventory.csv)

DQ_001

No NULLs found at the following columns:

inventory_date, room_type, total_rooms, out_of_service_rooms, available_rooms

Query:

```
SELECT
    inventory_date, room_type, total_rooms,
    out_of_service_rooms, available_rooms
FROM
    `Asteria_Boutique_Hotel.daily_room_inventory`
WHERE
    inventory_date IS NULL OR
    room_type IS NULL OR
    total_rooms IS NULL OR
    out_of_service_rooms IS NULL OR
    available_rooms IS NULL
```

DQ_002

No negative values found in the columns **total_rooms** & **available_rooms**.

Query:

```
SELECT
    inventory_date, total_rooms, available_rooms, out_of_service_rooms
FROM
    `Asteria_Boutique_Hotel.daily_room_inventory`
WHERE
    total_rooms < 0 OR
    available_rooms < 0 OR
    out_of_service_rooms < 0
```

DQ_003

No differences found on the calculation in **total_rooms** (**available_rooms** + **out_of_service_rooms**)

Query:

```
SELECT
    inventory_date,
    room_type,
    total_rooms,
    available_rooms,
    out_of_service_rooms
FROM
    `Asteria_Boutique_Hotel.daily_room_inventory`
WHERE
    total_rooms <> (available_rooms + out_of_service_rooms)
```

DQ_004

No empty cells found in the column room_type (STRING)

Query:

```
SELECT
    inventory_date,
    room_type,
FROM
    `Asteria_Boutique_Hotel.daily_room_inventory`
WHERE
    TRIM(room_type) = ""
```

DQ_005

No rows found with duplicate combinations between date (**inventory_date**) & room type (**room_type**).

Query:

```
SELECT
    inventory_date, room_type, count(*) AS times_found
FROM
    `Asteria_Boutique_Hotel.daily_room_inventory`
GROUP BY
    inventory_date, room_type
HAVING times_found > 1
```

DQ_006

Check of min/max date values as well the unique dates (**unique_dates**) with the rooms (**room_type**).

Query:

```
SELECT
    room_type,
    MIN(inventory_date) AS min_date,
    MAX(inventory_date) AS max_date,
    COUNT(DISTINCT(inventory_date)) AS unique_dates
FROM
    `Asteria_Boutique_Hotel.daily_room_inventory`
GROUP BY
    room_type
```

Job information		Results	Visualization	JSON	Execution details	Execution
Row	room_type	min_date	max_date	unique_dates		
1	Suite	2024-01-01	2025-12-31	731		
2	Single	2024-01-01	2025-12-31	731		
3	Double	2024-01-01	2025-12-31	731		

Data Cleaning

(hotel_bookings_raw.csv)

DC_001

Creation of a new table with the name "**hotel_bookings_clean**".

Query:

```
CREATE TABLE
    `Asteria_Boutique_Hotel.hotel_bookings_clean` AS
SELECT
    DISTINCT *
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_raw`
```

DC_002

Confirmation of rows in the new clean table. Expected rows: 3000.

Query:

```
SELECT
    COUNT(*) AS total_rows
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
```

> Query results

Job information		Results	Visu
Row	total_rows		
1	3000		

DC_003

Standardized **guest_country** codes to uppercase and removed leading and trailing spaces. Existing NULL values were preserved.

Query:

```
UPDATE
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
SET
  guest_country = UPPER(TRIM(guest_country))
WHERE
  TRUE
```

DC_004

Confirmed results in **guest_country** codes after the previous query.

Query:

```
SELECT
  DISTINCT guest_country
FROM
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
```

Job information		Results	Visu
Row	guest_country		
1	ES		
2	IT		
3	GR		
4	GB		
5	DE		
6	NL		
7	US		
8	FR		
9	null		

DC_005

Standardized **booking_channel** values to Booking.com, Direct, Expedia, and Travel Agency, resolving differences in capitalization and leading or trailing spaces.

Query:

```
UPDATE
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
SET
  booking_channel = CASE LOWER(TRIM(booking_channel))
    WHEN "booking.com" THEN "Booking.com"
    WHEN "direct" THEN "Direct"
    WHEN "expedia" THEN "Expedia"
    WHEN "travel agency" THEN "Travel Agency"
    ELSE booking_channel
  END
WHERE TRUE
```

DC_006

Confirmed results in **booking_channel** values after the previous query.

Query:

```
SELECT
  DISTINCT booking_channel
FROM
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
```

Job information		Results	Visualization
Row	booking_channel		
1	Booking.com		
2	Direct		
3	Expedia		
4	Travel Agency		

DC_007

Standardized **trip_purpose** values to Business and Leisure, resolving differences in capitalization and leading or trailing spaces.

Query:

```
UPDATE
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
SET trip_purpose = CASE LOWER(TRIM(trip_purpose))
  WHEN "business" THEN "Business"
  WHEN "leisure" THEN "Leisure"
  ELSE trip_purpose
END
WHERE TRUE
```

DC_008

Confirmed results in **trip_purpose** values after the previous query.

Query:

```
SELECT
  DISTINCT trip_purpose
FROM
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
```

Job information		Results	Visua
Row	trip_purpose		
1	Leisure		
2	Business		

DC_009

Replaced 15 invalid **review_score** values outside the 1–10 range with NULL. Valid scores and existing NULL values were preserved.

Query:

```
UPDATE
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
SET
  review_score = NULL
WHERE
  review_score < 1 OR review_score > 10
```

DC_010

Confirmation of the **review_score** column values. No values lower than 1 and higher than 10 were returned.

Query:

```
SELECT
  review_score
FROM
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
  review_score < 1 OR review_score > 10
```

DC_011

Corrected **room_total_eur** for 15 bookings where it was overstated by €25, using **booking_value_eur – extras_total_eur**

Query:

```
UPDATE
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
SET
  room_total_eur = ROUND(booking_value_eur - extras_total_eur, 2)
WHERE
  ROUND(room_total_eur + extras_total_eur - booking_value_eur, 2) = 25
```

DC_012

Confirmation of the previous query. No results as expected.

Query:

```
SELECT
  booking_id,
  room_total_eur,
  extras_total_eur,
  booking_value_eur
FROM
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
  ROUND(room_total_eur + extras_total_eur - booking_value_eur, 2) <> 0
```

Data Analysis

(hotel_bookings_clean.csv & daily_room_inventory.csv)

1. Out of all the bookings, how many were completed, cancelled, or resulted in a no-show?

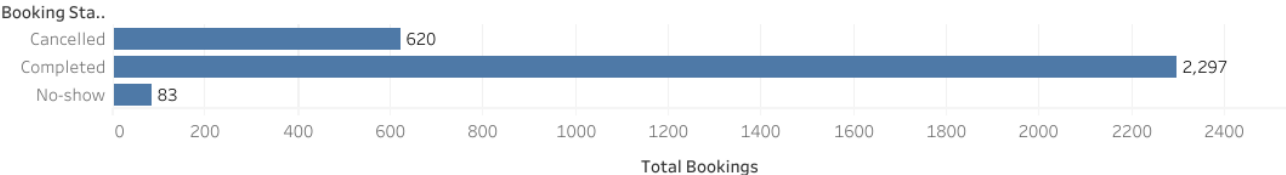
Query:

```
SELECT
  booking_status,
  COUNT(*) AS sum_booking_status,
  ROUND((COUNT(*) / 3000 * 100), 2) AS booking_status_pct
FROM
  `Asteria_Boutique_Hotel.hotel_bookings_clean`
GROUP BY
  booking_status
```

Results:

Job information		Results	Visualization	JSON	Execution
Row	booking_status	sum_booking_sta...	booking_status_pct		
1	Completed	2297	76.57		
2	Cancelled	620	20.67		
3	No-show	83	2.77		

Bookings by Status | 2024-2025
Completed, Cancelled & No-show • 3.000 bookings



Findings:

The key finding here is that 2,297 of the 3,000 bookings were completed, while 703—or 23.43%—were cancelled or resulted in a no-show. We do not know whether these rooms were resold, so we do not automatically classify this as lost revenue.

2. How many bookings did each channel generate, regardless of their final status?

Query:

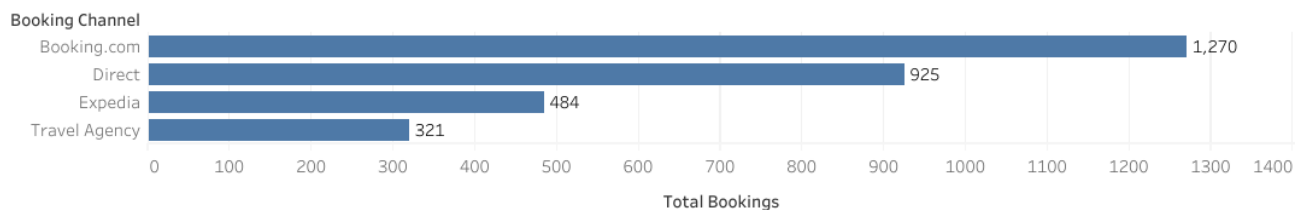
```
SELECT
    booking_channel,
    COUNT(*) AS sum_booking_channel,
    ROUND((COUNT(*) / 3000 * 100),2) AS booking_channel_pct
FROM
    `AsteriaBoutique_Hotel.hotel_bookings_clean`
GROUP BY
    booking_channel
```

Results:

Row	booking_channel	sum_booking_ch...	booking_channel...
1	Booking.com	1270	42.33
2	Direct	925	30.83
3	Expedia	484	16.13
4	Travel Agency	321	10.7

Bookings by Channel | 2024-2025

All booking statuses • 3,000 bookings



Findings:

Booking.com accounted for the largest share of bookings, with 1,270 bookings (42.33%), followed by direct bookings with 925 (30.83%). Overall, intermediaries—Booking.com, Expedia, and travel agencies—accounted for 69.17% of all bookings.

3. What happened to each channel's bookings?

Query:

```

SELECT
    booking_channel,
    booking_status,
    COUNT(*) AS specific_number
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
GROUP BY
    booking_channel,
    booking_status
ORDER BY
    booking_channel DESC

```

Results:

Row	booking_channel	booking_status	specific_number
1	Travel Agency	Completed	244
2	Travel Agency	Cancelled	71
3	Travel Agency	No-show	6
4	Expedia	Completed	381
5	Expedia	Cancelled	87
6	Expedia	No-show	16
7	Direct	Completed	704
8	Direct	Cancelled	191
9	Direct	No-show	30
10	Booking.com	Completed	968
11	Booking.com	Cancelled	271
12	Booking.com	No-show	31

Findings:

Completed bookings represented the majority of bookings across all channels. Booking.com recorded the highest number of completed bookings (968), followed by direct bookings (704), Expedia (381), and travel agencies (244). Booking.com also recorded the most cancellations (271), but it handled the largest overall booking volume. Cancellation rates are examined in the next section to compare channels relative to their total bookings.

4. How often are reservations from each channel cancelled or result in a no-show?**Query:**

```

SELECT
    booking_channel,
    COUNT(*) AS total_bookings_per_channel,
    ROUND((COUNTIF(booking_status = "Cancelled") / COUNT(*) * 100),2) AS cancelled_pct,
    ROUND((COUNTIF(booking_status = "No-show") / COUNT(*) * 100),2) AS no_show_pct
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
GROUP BY
    booking_channel

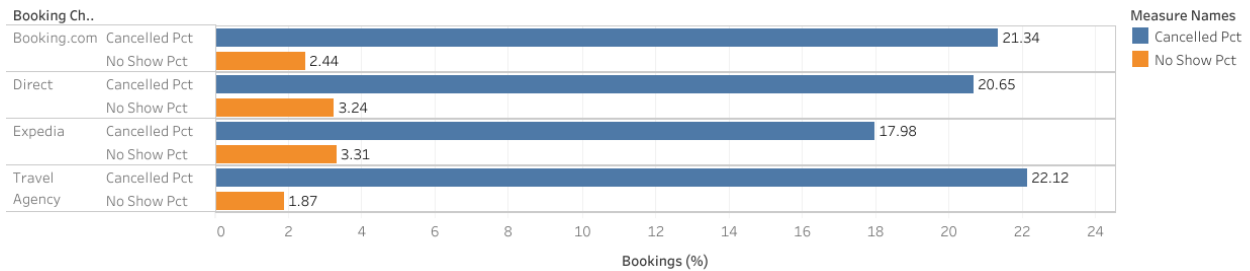
```

Results:

Row	booking_channel	total_bookings_p...	cancelled_pct	no_show_pct
1	Booking.com	1270	21.34	2.44
2	Direct	925	20.65	3.24
3	Expedia	484	17.98	3.31
4	Travel Agency	321	22.12	1.87

Cancellation & No-show Rates by Channel | 2024-2025

Percentage of each channel's total bookings



Findings:

Travel agencies recorded the highest cancellation rate (22.12%), followed by Booking.com (21.34%), direct bookings (20.65%), and Expedia (17.98%). Expedia had the highest no-show rate (3.31%), closely followed by direct bookings (3.24%), while travel agencies had the lowest (1.87%). These percentages are calculated within each channel, accounting for differences in total booking volume.

5. How does the number of completed bookings vary by arrival month? How do the corresponding months of 2024 and 2025 compare?

Query:

```
SELECT
    EXTRACT(YEAR FROM check_in) as arrival_year,
    EXTRACT(MONTH FROM check_in) as arrival_month,
    COUNT(*) completed_bookings
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    arrival_year, arrival_month
ORDER BY
    arrival_month
```

Results:

Row	arrival_year ▼	arrival_month ▼	completed_booki...
1	2024	1	94
2	2025	1	77
3	2025	2	71
4	2024	2	70
5	2025	3	81
6	2024	3	88
7	2024	4	68
8	2025	4	86
9	2024	5	69
10	2025	5	87
11	2025	6	129
12	2024	6	138
13	2024	7	120
14	2025	7	133
15	2024	8	151
16	2025	8	134
17	2025	9	126
18	2024	9	122
19	2024	10	74
20	2025	10	75
21	2024	11	87
22	2025	11	68
23	2024	12	81
24	2025	12	68

Findings:

Completed bookings showed a seasonal pattern in both years, with the highest monthly counts occurring from June to September. August was the busiest arrival month, recording 151 completed bookings in 2024 and 134 in 2025. Compared with 2024, April, May, July, and September recorded increases in 2025, while January, August, November, and December recorded declines. These results describe booking volumes by arrival month, rather than room occupancy.

6. Which room types have the highest number of completed bookings and the most overnight stays?

Query:

```
SELECT
    room_type,
    COUNT(*) AS total_bookings,
    SUM(DATE_DIFF(check_out, check_in, DAY)) AS sum_total_nights
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    room_type
ORDER BY
    room_type
```

Results:

Row	room_type	total_bookings	sum_total_nights
1	Double	1377	5361
2	Single	490	1930
3	Suite	430	1661

Findings:

Double rooms recorded the highest number of completed bookings (1,377) and room nights (5,361), followed by Single rooms with 490 bookings and 1,930 room nights, and Suites with 430 bookings and 1,661 room nights. Double rooms therefore contributed the largest volume of completed stays. However, these totals do not account for the number of available rooms in each category and should not be interpreted as occupancy rates.

7. How long does a complete stay last on average? Does the duration vary depending on the room type and the purpose of the trip?

Query 1:

```
SELECT
    ROUND(AVG(DATE_DIFF(check_out, check_in, DAY)),2) AS avg_overnights
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
```

Row	avg_overnights
1	3.9

Query 2:

```

SELECT
    room_type,
    trip_purpose,
    ROUND(AVG(DATE_DIFF(check_out, check_in, DAY)),2) AS avg_overnights
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    room_type, trip_purpose

```

Results:

Row	room_type	trip_purpose	avg_overnights
1	Single	Leisure	3.9
2	Double	Leisure	3.89
3	Suite	Leisure	3.88
4	Double	Business	3.89
5	Single	Business	4.26
6	Suite	Business	3.79

Findings:

Completed stays lasted an average of 3.90 nights. Leisure stays had similar average durations across all room types, ranging from 3.88 to 3.90 nights. Business stays showed greater variation: Single rooms recorded the longest average stay at 4.26 nights, followed by Double rooms at 3.89 nights and Suites at 3.79 nights.

8. How far in advance of arrival are bookings made? Does this vary by channel and final status?**Query 1:**

```

SELECT
    booking_channel,
    booking_status,
    ROUND(AVG(DATE_DIFF(check_in, booking_date, DAY)),2) AS avg_lead_time_days
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
GROUP BY
    booking_channel,
    booking_status

```

Results_1:

Row	booking_channel	booking_status	avg_lead_time_d...
1	Booking.com	Completed	92.32
2	Direct	Completed	91.62
3	Expedia	Completed	88.56
4	Travel Agency	Completed	92.89
5	Booking.com	Cancelled	88.36
6	Booking.com	No-show	100.97
7	Direct	Cancelled	89.51
8	Direct	No-show	88.53
9	Expedia	Cancelled	94.0
10	Expedia	No-show	92.75
11	Travel Agency	Cancelled	95.75
12	Travel Agency	No-show	122.33

Query 2:

```
SELECT  
    ROUND(AVG(DATE_DIFF(check_in, booking_date, DAY)),2) AS avg_lead_time_days  
FROM  
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
```

Results 2:

Row	avg_lead_time_d...
1	91.43

Findings:

Bookings were made an average of 91.43 days before the scheduled arrival date. For completed bookings, average lead times were similar across channels, ranging from 88.56 days for Expedia to 92.89 days for travel agencies. Cancelled bookings showed no consistent pattern of longer or shorter lead times compared with completed bookings. Travel agency no-shows had the longest average lead time (122.33 days), although this result is based on only six bookings and should be interpreted cautiously.

9. From which countries do the majority of completed bookings originate?

Query:

```
SELECT
    guest_country,
    COUNT(*) AS country_bookings
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    guest_country
ORDER BY
    country_bookings DESC
```

Results:

Row	guest_country	country_bookings
1	GR	640
2	GB	414
3	DE	340
4	FR	243
5	IT	221
6	US	158
7	NL	152
8	ES	124
9	<i>null</i>	5

Findings:

Greece accounted for the highest number of completed bookings (640), followed by the United Kingdom (414) and Germany (340). Together, these three countries represented 60.69% of all completed bookings. France and Italy followed with 243 and 221 bookings, respectively. The guest country was missing for five completed bookings.

10. How do business and leisure customers differ?

Query:

```
SELECT
    trip_purpose,
    COUNT(*) AS total_bookings,
    ROUND(AVG(DATE_DIFF(check_out, check_in, DAY)),2) AS avg_duration,
    ROUND(AVG(booking_value_eur),2) AS avg_booking_value_eur
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    trip_purpose
```

Results:

Row	trip_purpose	total_bookings	avg_duration	avg_booking_value_eur
1	Leisure	1950	3.89	507.55
2	Business	347	3.93	532.6

Findings:

Leisure travel accounted for the majority of completed bookings (1,950), compared with 347 business bookings. Average stay duration was similar for both groups, at 3.89 nights for leisure and 3.93 nights for business. Business bookings had a higher average booking value (€532.60) than leisure bookings (€507.55), a difference of €25.05 per booking.

11. How much revenue is associated with completed bookings?

Query:

```
SELECT
    SUM(room_total_eur) AS total_room_income,
    SUM(extras_total_eur) AS total_extras_income,
    (SUM(room_total_eur) + SUM(extras_total_eur)) AS total_income
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
```

Results:

Row	total_room_income	total_extras_income	total_income
1	1111762.34	62775	1174537.34

Findings:

Completed bookings generated a total booking value of €1,174,537.34. Room revenue contributed €1,111,762.34, while extras accounted for €62,775.00. Accommodation was therefore the main source of revenue. These amounts are before the deduction of booking commissions and operating expenses.

12. How do revenues vary by month and year?

Query:

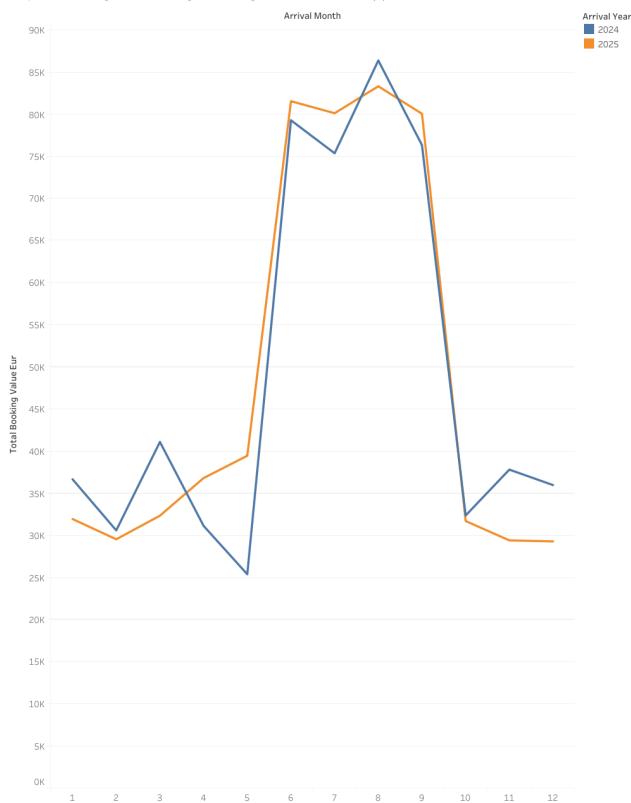
```
SELECT
    EXTRACT(YEAR FROM check_in) AS arrival_year,
    EXTRACT(MONTH FROM check_in) AS arrival_month,
    ROUND(SUM(booking_value_eur),2) AS total_booking_value_eur
FROM
    `Asteria-Boutique-Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    arrival_year, arrival_month
ORDER BY
    arrival_year, arrival_month
```

Results:

Row	arrival_year	arrival_month	total_booking_val...
1	2024	1	36687.62
2	2024	2	30625.17
3	2024	3	41121.75
4	2024	4	31159.03
5	2024	5	25419.93
6	2024	6	79304.67
7	2024	7	75386.55
8	2024	8	86399.53
9	2024	9	76354.65
10	2024	10	32395.06
11	2024	11	37839.17
12	2024	12	36002.57
13	2025	1	31970.87
14	2025	2	29569.61
15	2025	3	32370.12
16	2025	4	36838.35
17	2025	5	39480.93
18	2025	6	81565.05
19	2025	7	80142.2
20	2025	8	83338.01
21	2025	9	80074.85
22	2025	10	31734.0
23	2025	11	29438.95
24	2025	12	29318.7

Monthly Booking Value | 2024-2025

Completed Bookings • Full booking value assigned to arrival month (€)



Findings:

The total value of completed bookings showed a clear seasonal pattern, with the highest monthly values recorded from June to September in both years. August peaked at €86,399.53 in 2024 and €83,338.01 in 2025. Compared with 2024, June, July, and September recorded higher values in 2025, while August declined. These figures assign each booking's full value to its arrival month, rather than allocating revenue across the individual nights of the stay.

13. Which channels generate the most revenue, and how much remains after commissions?

Query:

```
SELECT
    booking_channel,
    ROUND(SUM(booking_value_eur),2) AS total_booking_value_eur,
    ROUND(SUM(commission_eur),2) AS total_commission,
    ROUND((ROUND(SUM(booking_value_eur),2) - ROUND(SUM(commission_eur),2)),2) AS
difference_without_commission
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    booking_channel
ORDER BY
    total_booking_value_eur DESC
```

Results:

Row	booking_channel	total_booking_value...	total_commission	difference_without_commission
1	Booking.com	488333.87	69279.2	419054.67
2	Direct	367242.12	0.0	367242.12
3	Expedia	192432.3	32656.91	159775.39
4	Travel Agency	126529.05	12009.92	114519.13

Findings:

Booking.com generated the highest value of completed bookings (€488,333.87), with €419,054.67 remaining after commissions. Direct bookings followed with €367,242.12 and no recorded commissions. Expedia contributed €159,775.39 after commissions, while travel agencies contributed €114,519.13. Booking.com therefore remained the largest revenue contributor after commission deductions. These remaining amounts do not represent profit, as other operating costs have not been deducted.

14.What is the average room revenue per sold night – ADR?

Query:

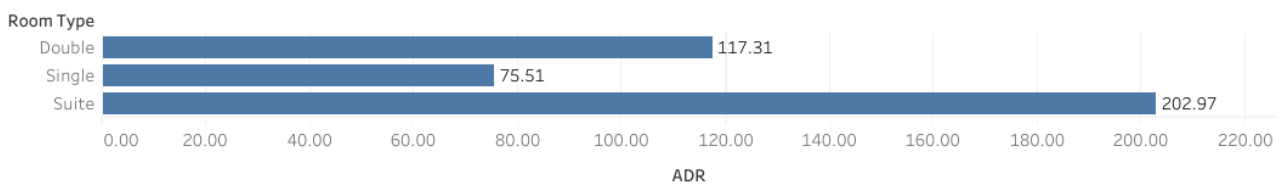
```
SELECT
    room_type,
    ROUND(SUM(room_total_eur),2) AS sum_room_total_eur,
    SUM(
DATE_DIFF(check_out, check_in, DAY)) AS total_nights,
    ROUND(SUM(room_total_eur) / SUM(
DATE_DIFF(check_out, check_in, DAY)),2) AS ADR
FROM
    `Asteria_Boutique_Hotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    room_type
```

Results:

Row	room_type	sum_room_total_eur	total_nights	ADR
1	Single	145724.91	1930	75.51
2	Double	628904.82	5361	117.31
3	Suite	337132.61	1661	202.97

Average Daily Rate by Room (Type) | 2024-2025

Completed Stays • Room revenue per sold room night (€)



Findings:

Suites recorded the highest average daily rate (ADR) at €202.97 per sold room night, followed by Double rooms at €117.31 and Single rooms at €75.51. Despite their higher ADR, Suites generated less total room revenue than Double rooms, which accounted for substantially more sold room nights (5,361 versus 1,661). ADR includes accommodation revenue only, excluding extras, and is calculated before commission deductions.

15. How do guests rate their stay?

Query:

```
SELECT
    room_type,
    ROUND(AVG(review_score),2) AS avg_review_score,
    COUNT(review_score) AS number_of_reviews
FROM
    `AsteriaBoutiqueHotel.hotel_bookings_clean`
WHERE
    booking_status = "Completed"
GROUP BY
    room_type
```

Results:

Row	room_ty...	avg_review_score	number_of_reviews
1	Single	8.53	351
2	Double	8.52	914
3	Suite	8.46	295

Findings:

Guest ratings were similar across all room types. Single rooms averaged 8.53 out of 10 based on 351 reviews, followed by Double rooms at 8.52 from 914 reviews and Suites at 8.46 from 295 reviews. The gap between the highest and lowest averages was only 0.07 points, indicating little variation in average ratings by room type. These results reflect completed bookings with an available review score.

16. What do we recommend to management based on the findings?

Based on the analysis, here's what I'd suggest management look at:

1. Review the cancellation policy

We're seeing about 20.67% of bookings cancelled. It might be worth testing some adjusted terms, or offering guests a clear choice between flexible and non-refundable rates. The goal is to get more reliable bookings without making the offer less attractive. We should then check how this affects both the cancellation rate and the number of stays that actually happen.

2. Push direct bookings a bit more, without dropping the existing channels

Direct bookings brought in €367,242.12 in completed value with zero commissions. A focused push on the hotel website could grow that share while still keeping the reach we get from the current partners. The real test will be whether the advertising spend is lower than the commissions we're saving.

3. Try some targeted offers in April, May and October

These months are quieter than the June–September peak. If we have the capacity, short-stay packages, added-value offers or selective discounts could help fill the gaps. Success should be measured by extra completed stays and net revenue after the promo costs, not just by how many bookings come in.

4. Look more closely at the business travel segment

Guests in Single rooms who are travelling for work stay the longest on average (4.26 nights). There's an opportunity here to test a simple business-oriented offer – maybe a comfortable workspace and solid internet. Guest feedback and actual demand should decide how far we go with any investment.

5. Time promotions around the booking lead times

The average booking is made 91.43 days in advance, so campaign planning roughly three months ahead of target periods makes sense. Timing and spend can then be fine-tuned based on how bookings are pacing and what's still available. A closer look at the full lead-time distribution would help sharpen this further.